

Review

Emerging Perspectives on Resource Tracking and Animal Movement Ecology

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Resource tracking, where animals increase energy gain by moving to track phenological variation in resources across space, is emerging as a fundamental attribute of animal movement ecology. However, a theoretical framework to understand when and where resource tracking should occur, and how resource tracking should lead to emergent ecological patterns, is lacking. We present a framework that unites concepts from optimal foraging theory and landscape ecology, which can be used to generate and test predictions on how resource dynamics shape animal movement across taxa, systems, and scales. Consideration of the interplay between animal movement and resource dynamics not only advances ecological understanding but can also guide biodiversity conservation in an era of global change.

Resource Tracking as a Building Block of Animal Movement Ecology

The spatiotemporal distribution of resources shapes ecological processes, from individual fitness to population dynamics and species interactions [1,2]. Animal movement is the primary link between these processes, allowing individuals to respond dynamically to ever changing, heterogeneous environments [3,4]. Research over the last decade has revealed how both the spatial and temporal configuration of resources, termed the ‘**resource landscape**’ (see [Glossary](#)), can profoundly underpin animal movement strategies [5]. More recently, the concept of ‘**resource tracking**’ describes how fine-scale movements in response to ‘**phenological variation**’ in resource availability can scale up to emergent movement patterns such as migration [6]. Drawing from the fields of behavioral and spatial ecology, the central tenet of resource tracking is that mobile consumers can benefit by moving to exploit phenological variation in resources across space. From ungulates tracking fleeting plant phenology [7–11] to whales tracking prey aggregations [12] to raptors tracking updrafts [13–15], resource tracking has been documented worldwide among a diverse range of systems and taxa ([Figure 1](#)).

Though resource tracking is fundamental to animal movement ecology [6], its theoretical foundation has not yet been articulated. Optimal foraging theory (OFT) and landscape ecology both offer classical ecological theory that underlie resource tracking. OFT is concerned with predicting the fine-scale behavioral decisions of animals based on the idea that animals make foraging decisions to maximize their energy intake and fitness [16–18]. Among other things, OFT has provided theoretical background on how animals make behavioral decisions under uncertainty [19], particularly when the environment varies in space and time [20]. Importantly, the manner in which animals forage under uncertainty sparked interest in the scale that animals perceive their environment [21,22]. Whereas OFT is rooted in classical ethology, which considers animal behavior at the scale of seconds to hours [23], resource tracking can be considered a manifestation of OFT played out over longer periods across large landscapes. Specifically, individual movement decisions intended to maximize energy intake in a variable environment lead to movement patterns at broader temporal scales [24].

Highlights

‘Resource tracking’ describes how animals move to exploit resource availability that changes across space and time, a behavior that should be beneficial and ubiquitous in many systems and taxa.

However, when and how animals track resources will depend on the spatiotemporal configuration of resources in their environment.

We present a unifying framework to quantify resource dynamics, which allows for testing predictions on emergent movement behaviors in different resource configurations.

Resource tracking has important consequences for individual fitness, population dynamics, community interactions, and ecosystem services.

The reliance of animals on phenological diversity in resource availability, and on their ability to move without restrictions to track resources, is a significant concern for biodiversity conservation with human-induced rapid environmental change.

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At the other end of the spatial scale continuum, landscape ecology is concerned with the causes and consequences of spatial heterogeneity within mosaics of habitat patches (i.e., the landscape scale) [25]. Concepts derived from landscape ecology offer insight into a range of ecological processes [26]. For instance, landscape ecology highlighted the idea of landscape connectivity, whereby an animal's use of a habitat patch is dependent on its distance from the animal, the biophysical nature along the way, and mobility of the animal [27,28]. This framework has informed our understanding of conservation biology [29] and dispersal and invasion ecology, among others [30]. While landscape ecology has classically considered resource configurations as static [25], particularly at subannual timescales (but see [31]), studies of resource tracking explicitly consider how the spatial patterning of resources changes through time, typically at finer temporal extents of weeks to months. As landscape ecology has offered ecologists a framework for consistently defining and quantifying spatial heterogeneity in resources [32], a similar framework is needed for characterizing and quantifying spatiotemporal heterogeneity that incorporates the critical element of time.

The concept of resource tracking provides a theoretical bridge between: (i) the temporally explicit, often spatially implicit, field of OFT focused on animal behavior; and (ii) the spatially explicit, often temporally implicit, field of landscape ecology focused on the physical environment. Additionally, resource tracking serves as a lens to unify these two fields across spatial scales to better understand the drivers of animal movement.

Biophysical Drivers of Resource Landscapes

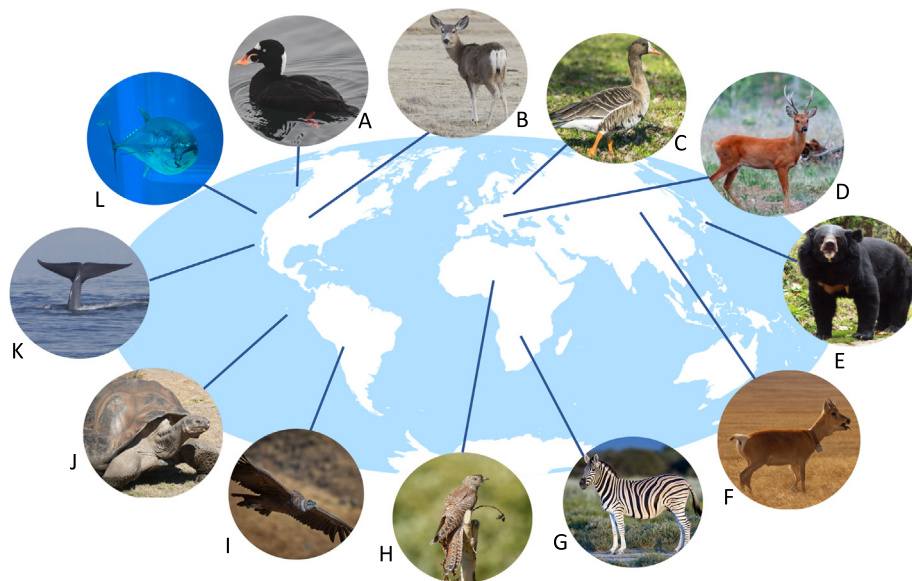
At the broadest spatial scale, Earth's tilt and orbit generate seasonal variation in photoperiod and climate. Geographic variation in seasonality sets the stage for biological variation at nested spatial scales (Figure 2A,B). For example, latitudinal variation in photoperiod, atmospheric patterns, and temperature generate corresponding gradients in the phenology of primary productivity, including phytoplankton blooms in aquatic and marine ecosystems [33,34] and spring green-up and flowering in terrestrial ecosystems [35–37]. Nested within latitudinal or continental gradients, geomorphic processes that generate topography and bathymetry shape finer scales of phenology. For example, elevation mediates the growing season for plants [35], such that low elevation sites within mountain ranges can green-up several months earlier than high elevation sites [7]. In riverine systems, water temperatures often vary longitudinally and generate corresponding gradients in the phenology of insects and fishes [38,39]. Within-site variation in phenology results from microclimate and hydrological variation. For example, equator-facing slopes experience more solar insolation, and depressions in topography (e.g., valley bottoms or marine canyons) accumulate cooler temperatures [40]. These topoclimatic mechanisms can generate phenological variation at small spatial scales.

While the physical environment exhibits heterogeneity at multiple spatial scales, the biological response to phenology is patterned at multiple organizational levels, including individual, population, and species. Coupling of environmental heterogeneity with variable biological responses act to prolong the overall availability of resources to consumers [41] (Figure 2C). Individuals may vary in responses to the same environmental cues (i.e., different reaction norms [42]), or individuals may have a common response but experience different microhabitats [39]. These individual-level differences cause broad phenological variation within a population. For example, in salmon, genetics give rise to individual-level variation in arrival at spawning sites [43,44]. The magnitude of individual-level variation in arrival date subsequently determines how long a population of salmon is available to predators [39]. Across salmon populations, variation in water temperature among spawning sites leads to population-level variation in average spawn timing [45]. This population-level variation across the landscape prolongs foraging opportunities for consumers that can

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	Species	Resource	Location	Driver of phenological variation	Movement strategy	Source
A	Surf scoter (<i>Melanitta perspicillata</i>)	Herring spawn	Alaska and Canada	Water temperature	Migratory	[103]
B	Mule deer (<i>Odocoileus hemionus</i>)	Herbaceous plants	Wyoming USA	Elevation, topography	Migratory	[7,10]
C	White-fronted goose (<i>Anser albifrons</i>)	Herbaceous plants	Europe and Russia	Temperature, snow cover	Migratory	[104]
D	Roe deer (<i>Capreolus capreolus</i>)	Herbaceous plants	Europe	Microclimate	Resident	[105]
E	Asiatic black bear (<i>Ursus thibetanus</i>)	Tree fruit	Japan	Elevation	Resident	[106]
F	Mongolian gazelle (<i>Procapra gutturosa</i>)	Herbaceous plants	Central Asia	Precipitation	Nomadic	[107]
G	Plains zebra (<i>Equus quagga</i>)	Herbaceous plants	Botswana	Precipitation	Migratory	[108]
H	Common cuckoo (<i>Cuculus canorus</i>)	Invertebrates	West Africa and Europe	Temperature, precipitation	Migratory	[109]
I	Andean condor (<i>Vultur gryphus</i>)	Wind thermal	Andes	Temperature	Resident	[13]
J	Galapagos tortoise (<i>Chelonoidis nigra</i>)	Herbaceous plants	Galapagos Archipelago	Elevation	Migratory	[51]
K	Blue whale (<i>Balaenoptera musculus</i>)	Krill	Eastern North Pacific	Wind-driven upwelling	Migratory	[12]
L	Pacific bluefin tuna (<i>Thunnus orientalis</i>)	Primary production	Eastern North Pacific	Wind-driven upwelling	Migratory	[53, 110]

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Figure 1. Resource Tracking Occurs Worldwide in Response to Phenological Variation in Resource Availability Driven by Physical and Biological Heterogeneity. See [7,10,12,13,51,53,103–110].

Glossary

Abundance: the quantity of resources at a given point in space and time.

Constancy: the inverse of variability (i.e., constancy increases as variability decreases).

Contingency: the degree of temporal autocorrelation for a variable of interest.

Ephemerality: the length of time that resources are available at a given location.

Phenological variation: variation in timing expressed across space or levels of biological organization.

Predictability: the variability and autocorrelation in resources across time at a given location (see constancy and contingency).

Resource landscape: the spatial and temporal configuration of resources.

Resource tracking: the movement of mobile consumers to follow phenological variation in resources across space.

Resource wave: an ordered gradient in resource phenology across space, typically generated by natural physical gradients such as elevation or latitude.

Semivariogram: a geostatistical tool, also called a variogram, that measures the degree of autocorrelation in a variable at different spatial scales. Semivariograms of the autocorrelation in the date of peak resource availability at different spatial scales can be used to quantify resource landscapes.

Spatial configuration: the spatial pattern of patch-level resource characteristics on a landscape.

Timing: the time when resources are available or peak in quality or quantity.

Variance: the degree of variation in patch-level characteristics distributed on a landscape.

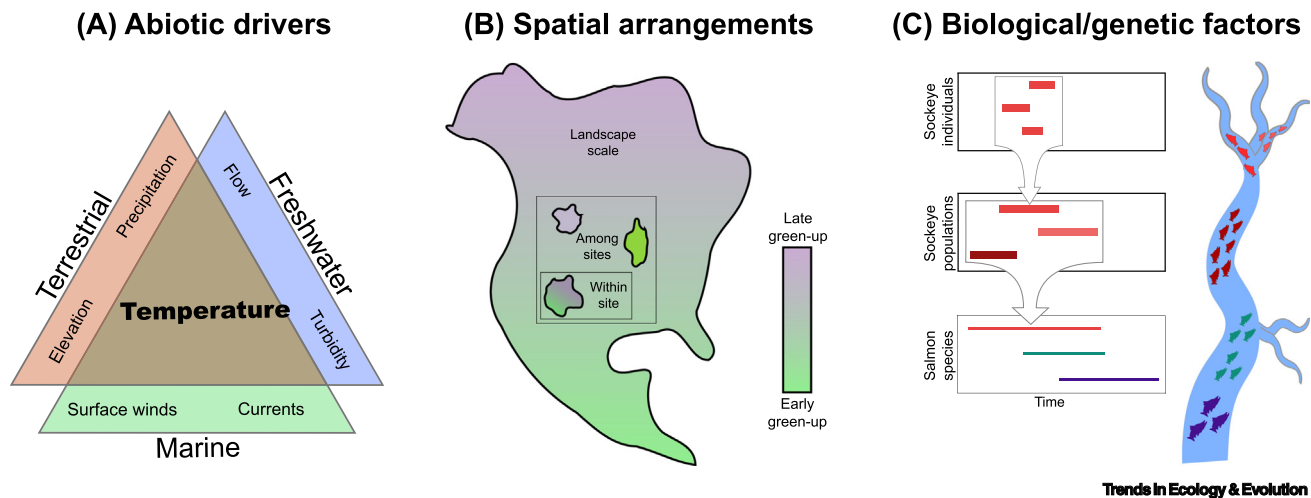


Figure 2. Phenological Variation at Multiple Levels of Biological and Spatial Organization Shapes Resource Landscapes and Leads to Increased Resource Availability for Consumers. (A) Physical drivers interact at different scales to create complex patterns of abiotic variation, which provide a template for the evolution and behavior of living things. (B) Intrasite, intersite, and landscape-level phenological variation in plant biomass due to topographic diversity lead to prolonged resource availability for migratory ungulates [37]. (C) Individual-, population-, and species-level phenological variation in salmon lead to prolonged resource availability for foraging brown bears [39].

move among spawning sites [46]. Furthermore, many water bodies have multiple species of salmon and these species often have different relationships between water temperature and spawning phenology. Together, this variation can increase the duration of salmon foraging opportunities for consumers such that spawning salmon can be available for several months in some basins, though individuals are only present and spawning for days or weeks [47].

Characterizing Resource Landscapes and Their Consequences for Animal Movement

The resource landscape determines the availability of energy to mobile consumers, which in turn shapes animal movement. The spatial scale of the resource landscape depends on the ecological process, time period, and mobility or activity of the study species in question [48]. While most studies of resource tracking focus on food resources, animals also track non-food resources that reduce energetic output or increase survival. For example, soaring birds track seasonal emergence of wind thermals [15]; migrating waterfowl and ungulates track snow or ice melt [49,50]; ectotherms such as tuna, salmon, and tortoises track thermal landscapes over large distances [51–53]; and vertically migrating fish track seasonal changes in light environments to avoid predation [54]. Given the incredibly broad diversity of resource landscapes on Earth, a generalizable framework for their characterization and quantification is necessary in order to test hypotheses across systems.

We expand upon Mueller *et al.* [24] to characterize the resource landscape according to six axes (Figure 3). **Abundance, timing, ephemerality, and predictability** are defined as patch-level characteristics. The **spatial configuration** and **variance** of patch-level characteristics constitute emergent landscape-level characteristics. All of these axes interact with one another to produce resource dynamics that influence animal movement. By applying these axes, several predictions about resource tracking and animal movement emerge that often challenge traditional conceptions.

For example, two patches that differ in total resource abundance might be classified as low- or high-quality habitat, with the assumption being that animals will use the high-quality (i.e., higher

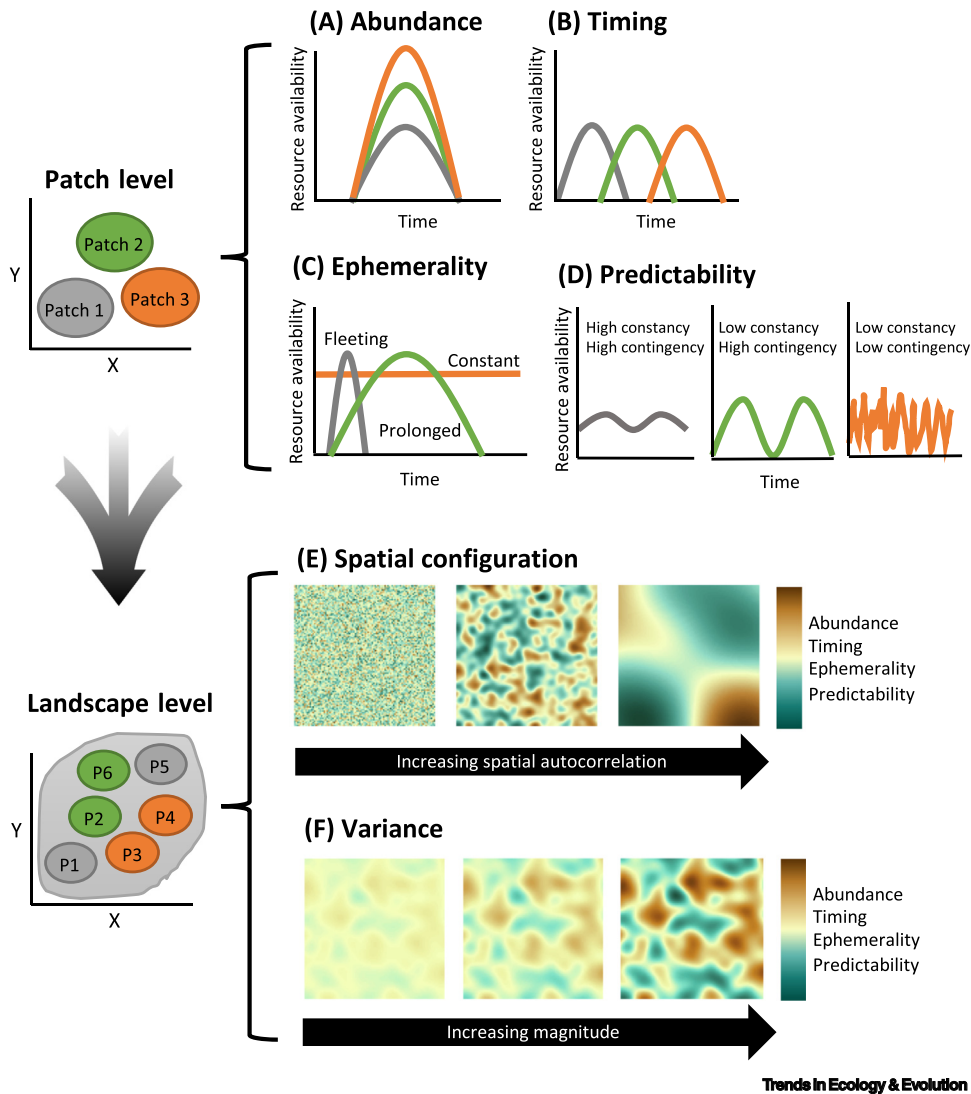
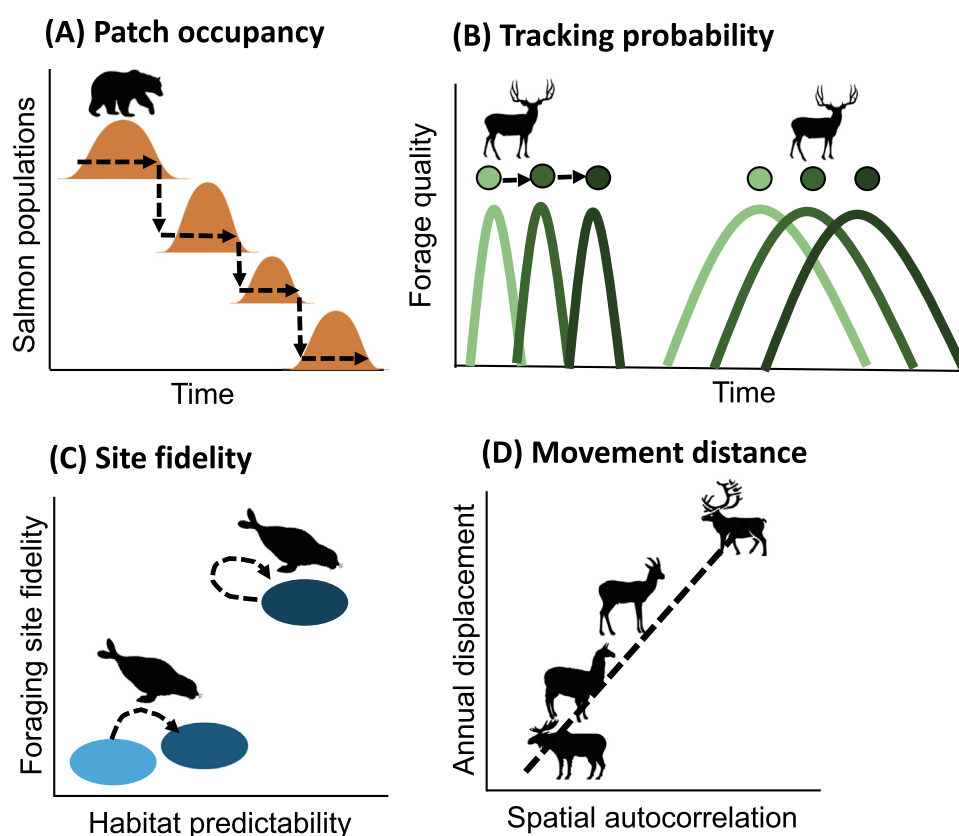


Figure 3. The Six Axes of the Resource Landscape. These include abundance (A), timing (B), ephemerality (C), predictability (D), spatial configuration (E), and variance (F). (A–D) Patch-level characteristics that, when configured on a landscape (E, F), influence the total availability of resources to a consumer. Resource abundance characterizes the amount of resources available at a given location and time (A). Timing constitutes when in the annual cycle a resource is available or peaks in quality or quantity (B). Ephemerality constitutes how long resources are available across time, where at one end of the spectrum a resource can be available during a narrow window of time (i.e., fleeting) and at the other end of the spectrum the resource may be available all of the time (i.e., constant; C). The concepts of resource abundance, timing, and ephemerality are complementary, in that resource abundance can be thought of as the height of a curve characterizing a resource pulse (A), timing is the position of the curve (B), and ephemerality is the width of the curve (C). Finally, predictability is dependent on how much the resource varies across time (i.e., constancy) and the temporal autocorrelation in resource variation (i.e., contingency). Resources may be predictable in that they change very little through time (high constancy) or that they change in a periodic fashion (high contingency). Decreases in constancy and contingency lead to decreased predictability, with environments that have highly variable resource fluctuations with no pattern or periodicity representing the most unpredictable environments. Patch-level characteristics may be compared across patches, which scale up to determine their landscape-level spatial configuration (E) and variance (F) in the variable of interest (e.g., abundance or timing of peak resource availability). The degree of spatial autocorrelation and magnitude of phenological variation in resources can be quantified with semivariograms (Box 1).

resource abundance) habitat more frequently than the lower-quality habitat (i.e., lower resource abundance), or that high-quality habitats will support a greater density of animals [55] (Figure 3A). However, if the timing of peak abundance across patches varies, then moving across patches would extend the time window when resources are available (Figure 3B). Thus, habitats with relatively low resource abundance may contribute disproportionately to consumers if their phenological patterns provide temporally unique foraging opportunities (Figure 4A).

Ephemerality describes how long resources are accessible at a given point on the landscape (Figure 3C). For example, fleeting resources that are only available for a short period of time are likely to favor close tracking of peak resource availability, whereas resources that are more



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Figure 4. The Six Interacting Axes of Resource Landscapes Lead to Emergent Predictions on the Consequences of Resource Tracking as an Underlying Mechanism in Animal Movement Ecology. (A) Lower abundance resource patches, traditionally viewed as lower quality habitat, will be disproportionately important if they exhibit unique phenologies. Example: brown bears (*Ursus arctos*) rely on the phenological diversity provided by small populations of salmon that spawn early in the year and variation in salmon spawning phenology explains patch occupancy of bears across the landscape [39,46]. (B) The probability of moving to track resources will be greatest in environments with fleeting resources. Example: mule deer (*Odocoileus hemionus*) more closely track the progression of springtime green-up when green-up rates are more rapid [7]. (C) Site fidelity will be greatest in environments with high constancy and contingency. Example: northern elephant seals (*Mirounga angustirostris*) with high foraging site fidelity forage in habitats with high long-term resource predictability, whereas seals that are not site faithful forage in unpredictable habitats [111]. (D) Animals will move greater distances with increasing spatial autocorrelation in resources in order to track variability in resources. Example: greater broad-scale variability in plant forage corresponds to larger annual displacements across moose (*Alces alces*), guanacos (*Lama guanicoe*), Mongolian gazelles (*Procapra gutturosa*), and caribou (*Rangifer tarandus*), following a continuum from range residency to nomadic to migratory behavior [5].

prolonged might still facilitate some tracking, but deviating from peak resource availability is less costly [7] (Figure 4B). Resource tracking is less likely in places with constant resource availability [5]. However, resource tracking is beneficial if the resource is constant but of low quality, or animals move to track an alternative resource pulse available elsewhere (e.g., predators moving to intercept migrant prey [56]).

Resource predictability can be described in terms of temporal ‘constancy’ and ‘contingency’ (Figure 3D) [57]. Greater predictability can occur through either little variation in a resource through time (i.e., high temporal constancy) or periodic variation in the resources through time (i.e., high temporal contingency) [58]. A seasonal environment is characterized by low constancy but high contingency, whereas a completely unpredictable landscape would have low constancy and low contingency (Figure 3D) [24]. More repeatable movement behaviors and greater fidelity to certain patches are more likely to emerge in patches with greater predictability [24,59] (Figure 4D). Likewise, moderate constancy and contingency, characterized by predictable peaks in resources that vary in absolute abundance across space and time, may result in animals that rely on memory or tracking of long-term averages [12].

Heterogeneity in patch-level characteristics leads to emergent properties at the landscape level that profoundly shape the broad-scale movement patterns of animals [5,24] (Figure 3E,F). Environments can vary in the spatial configuration of resource phenology, from a completely random distribution, to patches of areas with similar phenology, to ordered gradients (Figure 3E, Box 1). Increasing spatial autocorrelation should favor longer distance animal movements to track spatial heterogeneity in resource phenology [5–7,24] (Figure 4D). When resource phenology is highly ordered across space (i.e., is highly spatially autocorrelated), it can be described as a ‘resource wave’ [7]. Tracking of resource waves, which results in directional and long-distance movement such as migration, is a specific case of resource tracking and is prevalent in systems with wave-like resource phenology (Box 1) [7,60]. The total variance in patch-level characteristics in a landscape can also vary substantially, ranging from homogeneous to highly variable (Figure 3F). Increasing variance should also favor resource tracking to exploit phenological variation in the landscape.

Biological Constraints on Resource Tracking

Beyond the characteristics of the resource landscape itself, several constraints will influence resource tracking, including: cognitive capacity, physiology, niche breadth, trophic position, and life-history strategy.

Tracking phenological variation in resources requires that animals can detect and respond to changes in resource quality over time both within and outside the immediate location they currently reside. Animals can use a number of strategies to do this, depending on their cognitive ability to sense and respond to information and their ability to move [3]. First, highly mobile animals can use patch sampling techniques whereby they visit a number of adjacent habitat patches relatively quickly [61]. Second, animals can use their perceptual abilities to collect information on the state of adjacent habitat patches [8,62], a classic example being gradient-following in advection [63]. Thus, animals with a larger perceptual range should be better able to detect and track resources [63]. Third, animals can observe the behavior of conspecifics (i.e., collecting public information) to determine habitat quality in adjacent patches [19,64]. Finally, cognitive mechanisms such as memory capacity [65] can improve efficiency in tracking resources by remembering the spatial progression of phenology or where and when a resource typically peaks in quality or quantity [12,66,67]. Such characteristics facilitate successful resource tracking by collecting information about when to leave a patch and which patch to visit next.

Box 1. A Generalizable Framework to Quantify Spatial Configuration and Variance in Resource Landscapes

Semivariograms are a tool from geostatistics that examine the degree of autocorrelation between a variable of interest at different spatial scales [101]. Spatial and temporal resource dynamics can be examined using variograms by modeling the spatial autocorrelation of the date of peak resource availability at different scales (Figure I). If there is no spatial structure in peak resources (Figure IA), then the semivariance plotted across different distance lags will be a flat line (Figure IE). The patch size of areas that peak in resource availability at similar times increase with spatial autocorrelation (Figure IB,C), resulting in increasing semivariance values that plateau at larger distance lags (Figure IF,G). Wave-like resources are characterized by resource phenology that progresses sequentially across large landscapes (Figure ID). In an environment with a resource wave, the difference in the date of resource peaks increases with greater distances between points (Figure ID), resulting in a continued increase in the semivariance at larger distance lags (Figure IH). Several key metrics can be extracted from the semivariogram, including the landscape-level variance in phenology, or strength of the wave (i.e., the maximum semivariance, also called the sill; horizontal dark grey broken line in Figure IG) and the length of the wave (i.e., the distance lag where the maximum semivariance is reached, also called the range; vertical light grey broken line in Figure IG). It is useful to fit a flexible model to the empirical variogram (i.e., using a Matérn semivariogram), and from this model the sill and the range are estimated [102]. If animals move to track resource phenology, then the length of the wave can predict how far on average animals should move.

Key Questions to Consider When Quantifying Resource Waves

What resource is the animal tracking? Are there multiple resources (i.e., generalist species) or is there one key resource (i.e., specialist species)?

What phenological stage is most critical (e.g., flowering, fruiting, mature seeds, intermediate growth stages, maximum abundance, tightest spatial aggregations)?

Over what temporal scale does the resource vary? Over what spatial scale?

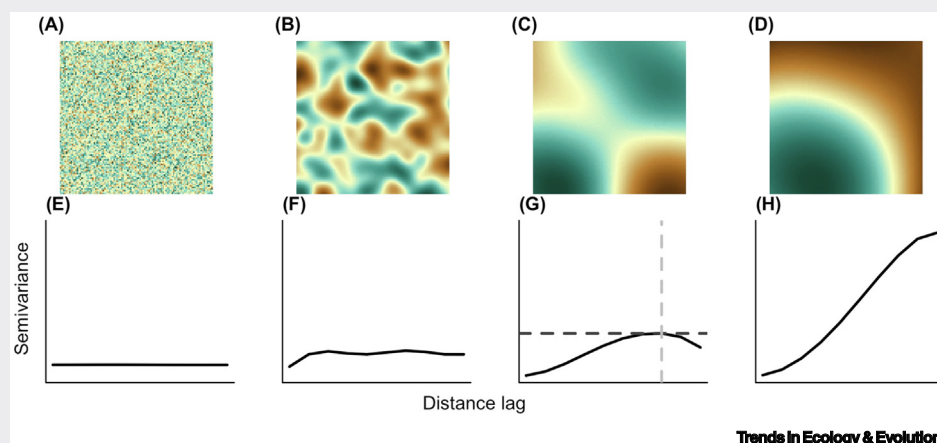


Figure I. An Illustration of How Semivariograms Can Be Used to Quantify the Wave-like Nature of Resource Phenology. (A–D) Simulated landscapes with varying degrees of spatial autocorrelation in the date of peak resource availability (the dark brown pixels represent earliest date of peak resource availability and the dark teal pixels represent the latest peak in resource availability). (E–H) The corresponding semivariograms for each simulated landscape, which measures the squared difference in the date of peak resource availability at different spatial lags (i.e., the semivariance). The maximum semivariance value can be interpreted as the strength of the wave (i.e., the sill; horizontal dark grey broken line in G), while the distance lag where the maximum semivariance is reached can be interpreted as the length of the wave (i.e., the range; vertical light grey broken line in G; [96]).

Whether or not a species engages in resource tracking may also be dependent on their physiology and niche breadth. Species that have a high physical tolerance for starvation, which enables them to cope with extended periods of food scarcity, may be less likely to move to follow food [68,69]. Similarly, a species' cost of transport, or locomotion capacity, will constrain resource tracking [13]. In addition, specialists tend to have narrower habitat and abiotic requirements or tolerances compared with generalists, which leads to two important predictions relevant to

resource tracking. First, compared with generalists, specialists perform well in their optimal environments or consuming their optimal prey, but at the expense of their performance with other environments or prey items [70,71]. It is therefore expected that if the fitness of a given habitat or prey varies in space and time, the specialized species would be more likely to track those resources. Second, and relatedly, generalist species are predicted to switch habitats or prey as profitability or availability changes over time [72]. Thus, when alternative resources exist, generalists may be less likely to move to track resources.

Depending on a species' trophic position, ecological interactions such as competition or predation risk can further mediate the degree to which an animal is able to track resources. For example, competition among conspecifics has been proposed as an explanation for why some animals 'jump' rather than 'surf' resource waves during migration, allowing them to arrive earlier and secure resources at breeding sites [73]. Resource tracking in migrating herbivores has also been shown to be increasingly constrained by both natural and human predators [50,74]. Finally, an animal's propensity to track resources may be shaped by its life-history strategy. For example, many migratory species are time-constrained by the need to reach their breeding range to establish a territory or arrive on breeding grounds in time for young to benefit from peak forage quality [73,75,76]. In many cases, animals must face trade-offs between maximizing foraging opportunities and meeting other life-history requirements.

Ecological Consequences of Resource Tracking

Resource tracking allows individuals to gain access to high-quality resources for a longer period of time than they would staying in place. Not surprisingly, new research is documenting the fitness benefits of tracking phenological variation in resources. Recent empirical work shows that bears that track spawning salmon increase their consumption over the year [47,77] and elk (*Cervus canadensis*) that better track resource waves of high-quality forage gain more fat during spring than their nonmigratory counterparts [11]. These fitness benefits allow populations to grow to numbers that exceed expected densities if they remained in the same location over time (*sensu* [78]).

Resource tracking not only has important consequences at the individual and population level, but it further impacts a range of ecosystem processes. The spatial configuration of the resource landscape and the degree to which animals track resources influence the distances animals move across their landscapes and their broader-scale movement strategies such as migration, residency, and nomadism [5,24,59], leading to higher-order effects. The long-distance movements of animals cause a myriad of ecological consequences by coupling remote ecosystems through both transport (e.g., propagule dispersal, nutrient deposition) and trophic services (e.g., herbivory, predator–prey coupling) [79]. The movement patterns of animals can also dramatically alter infectious disease risk [80]. For instance, the spread of avian influenza has been linked to the movement rates and distances in several waterfowl species [81]. Further, resource tracking in migratory elk mediates the link between annual environmental fluctuations and the distribution of brucellosis transmission risk to elk and livestock [82]. Thus, understanding how resource landscapes influence animal movements can facilitate predictions of how zoonotic diseases may propagate under different landscape conditions [83].

Conservation Implications of Resource Tracking in the Anthropocene

Increasingly, researchers are documenting that population performance of mobile consumers is linked to the ability of animals to move across phenological gradients [11]. Thus, alterations of natural landscapes and barriers that impede movement are likely to diminish the fitness benefits of resource tracking and represent a pressing conservation challenge [84,85]. A recent synthesis

found that animal movements were less expansive in habitats with a high human footprint, a result likely caused by both truncated movements and loss of long-distance movements such as migration [86]. Shorter movements suggest that human development may be impeding the ability of animals to exploit the full suite of phenological gradients previously available to them. Moreover, development has been shown to alter the pace of animal movement, which holds potential to cause phenological mismatches. For example, migrating mule deer (*Odocoileus hemionus*) speed up and spend less time in stopovers when moving along segments of migration corridors impacted by energy development [87] and these altered movement patterns caused a decrease in resource tracking as development increased over time [88]. As resource tracking studies proliferate, we will gain a more complete understanding of the ecological impacts of barriers on the free movement of animals.

Even in the absence of anthropogenic barriers, climate change threatens resource tracking by altering resource landscapes. For example, climate change is synchronizing biological and physical processes across marine and terrestrial ecosystems in North America [89], including synchronizing spring phenology across elevational [90] and latitudinal gradients [36]. Increased synchronization of phenology reduces the temporal availability of key resources, which diminishes the foraging benefit of migration for species that track fleeting resources [91,92]. Additionally, resource synchronization has potential to disrupt higher-level ecological processes. For example, increased synchronization in the timing of salmon spawning and elderberry availability during anomalously warm years resulted in Kodiak brown bears switching from consuming salmon to primarily consuming elderberries [93]. As a result, the coupling of terrestrial and freshwater systems facilitated by brown bears that forage in streams was disrupted [94]. Thus, altered resource landscapes and synchronization of resource phenology due to climate change can impact individual resource gain, alter food web dynamics, and disrupt nutrient flow between systems.

Resource tracking by mobile consumers is a complex behavior, which for some species may require experience, knowledge, memory, or other behavioral adaptations (see Biological Constraints on Resource Tracking). Preserving such cultural knowledge is emerging as an important consideration in conservation [95]. For example, Jesmer *et al.* [96] showed that reintroduced bighorn sheep (*Ovis canadensis*) and re-established moose (*Alces alces*) populations did not track resources well immediately following translocation. But their ability to track resources increased over multiple generations, as did their propensity to migrate, providing evidence that, at least in mammals, the ability to track resources over large landscapes must be learned. Studies that connect resource tracking to social or experiential learning, knowledge, and memory are still in their infancy. These studies, nonetheless, suggest that we ought to consider animal habitat as an integration of the resource landscape occupied by an individual in combination with the knowledge the animal has regarding how to exploit it. Successfully maintaining demographic benefits of resource tracking (and presumably, viable populations) then must require that habitats are kept intact and that the individuals or groups who possess the necessary knowledge are also conserved [95].

Concluding Remarks and Future Directions

Resource tracking provides an important mechanistic link between resource dynamics, fine-scale movement decisions, and emergent large-scale movement patterns. Understanding how resource landscapes shape animal movement is important for the study of animal ecology as well as for targeting effective conservation strategies in an era of human-induced rapid environmental change [97]. While resource tracking appears to be a widespread phenomenon for which ecological research is rapidly growing, it is still studied in a relatively narrow subset of systems. Advances in animal-borne telemetry and earth observation increasingly offer new

Outstanding Questions

What cues or navigational mechanisms enable resource tracking over long distances?

What are the eco-evolutionary outcomes of variation in resource waves and resulting resource tracking?

At what spatial or temporal extents does movement to track resources become nonoptimal?

How do intra- or interspecific interactions modify or constrain resource tracking?

What are the causes of intraspecific variation in resource tracking?

Is resource tracking experientially or socially learned?

How will climate change affect resource landscapes and animals' ability to track resources?

opportunities that should be pursued for investigation into animal responses to spatiotemporal variation in resources in previously understudied taxa and systems.

In addition, research on resource tracking, which has mainly been studied from an individual- or population-based movement ecology perspective, is ripe for integration with community ecology. For example, given emerging evidence of predation risk constraints on resource tracking (see Biological Constraints on Resource Tracking), there is an opportunity to unite the resource tracking concept with that of the landscape of fear [98]. Both lines of research independently seek to explain animal movement and resource selection, however, it is well recognized that animals must make trade-offs between both energetic or foraging opportunities and predation or competition risk. We foresee future studies integrating such bottom-up and top-down processes holistically (see Outstanding Questions). In addition, climate change can alter resource landscapes in ways that may restructure community dynamics through spatial [99] or phenological [100] community reassembly. For example, changes in resource waves may affect the movements of migratory herbivores, in turn affecting interactions with their predators through predator–prey migratory coupling [56]. Such research integrating resource dynamics, animal movement ecology, and community ecology will be critical to understand how climate- and human-driven environmental change affect individuals, species, communities, and ecosystems worldwide.

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